CLOUD SECURITY AND ARCHITECTURE

CSA Unit 4 Assignment
AWS EC2 Windows instance setup

Setup report | 28 October 2026

1. Objective

This assignment documents how to launch a Windows Server instance on Amazon EC2 and connect to it with Remote Desktop. The project guide names the services, the five setup steps, and the RDP path through an Elastic IP and a security group. The Administrator password is not recorded in this report.

The AWS CLI on this computer has no credentials, so a live instance was not launched. There is no public IP, instance ID, or console screenshot from an AWS account. The connection host below is therefore described by role, not by an invented address.

2. Services

Amazon VPC, a public subnet and its route table, a security group, a Windows Server EC2 instance, an Elastic IP, and an RDP client on the user laptop.

3. Setup steps

1. Create a VPC, attach an internet gateway, and add a public subnet whose route table sends 0.0.0.0/0 to that gateway.
2. Create a security group for the instance. Allow inbound TCP 3389 so Remote Desktop can reach Windows. Leave the private key and the Windows password out of the security group and out of the repository.
3. Launch a Windows Server EC2 instance in the public subnet, attach the security group, and select a key pair. The key pair is used later only to decrypt the Administrator password.
4. Allocate an Elastic IP and associate it with the instance so the RDP address stays stable.
5. In the EC2 console, open Connect and choose the RDP client. Decrypt the password with the private key on the local machine, then connect to the Elastic IP on port 3389 as Administrator. Do not copy that password into the report, the repository, or a screenshot.

4. Security configuration

The guide’s security control is the instance security group. RDP is the only inbound service the setup calls for. The password and the key pair stay outside this repository.

Control	Setting
Security group inbound	TCP 3389 (RDP), as required to connect to Windows.
Security group outbound	Default outbound rule so the instance can respond.
Network placement	Instance in a public subnet, with a route to an internet gateway.
Address	Elastic IP associated with the instance.
Identity	Key pair decrypts the Administrator password in the console. The private key is not stored here.
Secret handling	The decrypted password is excluded from this report and from the repository.

5. Connection details

These are the connection fields from the setup. The password cell is intentionally empty of any secret.

Field	Value
Protocol	Remote Desktop Protocol (RDP)
Port	3389
Username	Administrator
Password	Excluded. Decrypt it in the EC2 console and do not save it in this report.
Host	Elastic IP of the Windows instance. None was assigned, because no instance was launched.
Client	Remote Desktop client on the user laptop.

6. Diagram

Figure 1 follows the path written in the project guide: the laptop connects with RDP to the Elastic IP, the security group allows TCP 3389, and the instance sits in a public subnet inside the VPC.

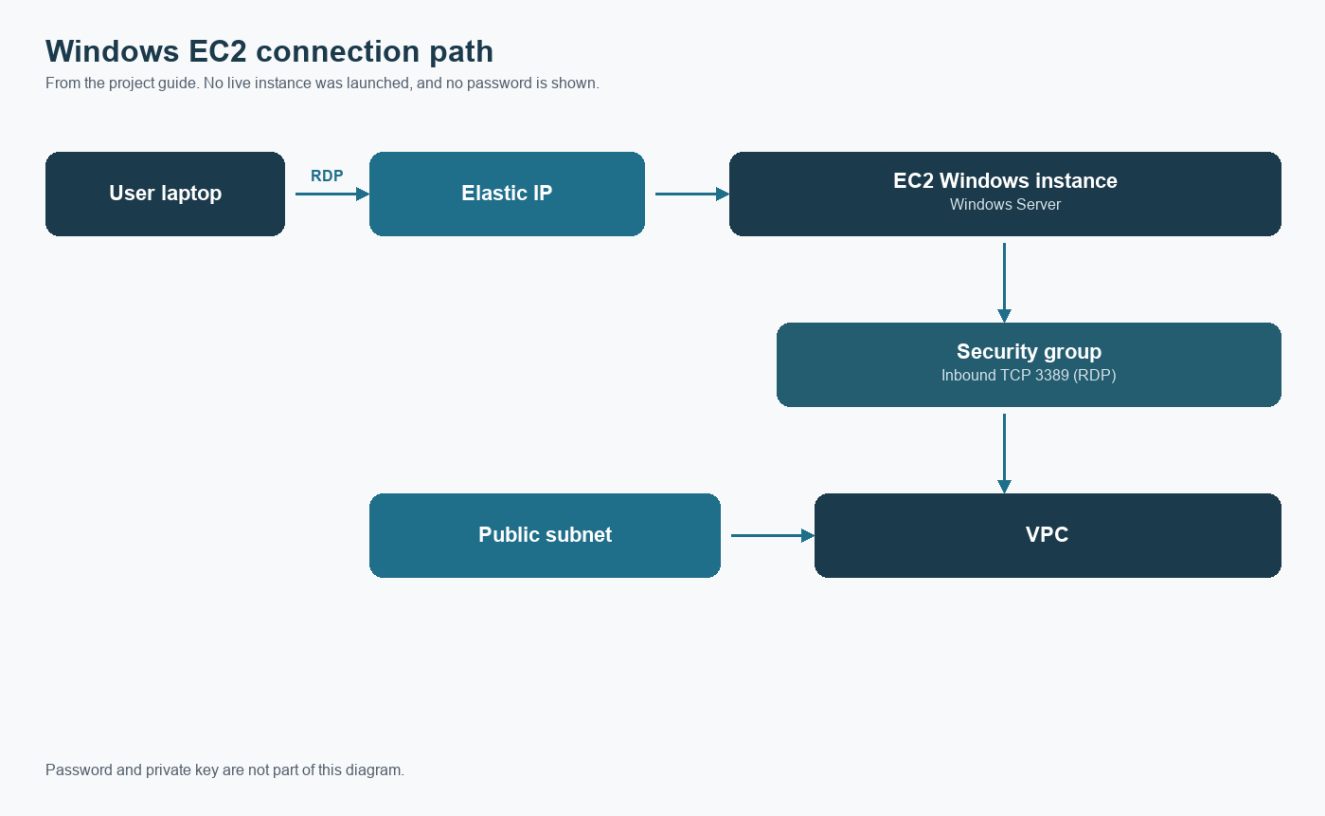

Figure 1. Architecture from the project guide. This is not an AWS console screenshot.

7. Conclusion

The Windows EC2 setup is a public-subnet instance reached by RDP on port 3389 through an Elastic IP. The security group is the control that permits that port, and the key pair is used only to decrypt the Administrator password. That password is not included here. A live launch, and screenshots of the instance, security group, and RDP session, still need an AWS account signed in on this computer.